

An MFEM HDG Solver for Steady Hypersonic Viscous Flow: Physics, Discretization, and Implementation

`myapps/hdg_navierstokes` — replication of the Exasim `nsmach8-baseline` run

July 2026

Abstract

This document describes the hybridizable discontinuous Galerkin (HDG) solver in `myapps/hdg_navierstokes`, an MFEM-based reimplementation of the Exasim `nsmach8-baseline` case: steady laminar flow of a perfect gas at $M_\infty = 8.03$, $\text{Re} = 1.835 \times 10^5$ over an isothermal half-cylinder. The solver reproduces Exasim’s discretization to residual-level parity (the Exasim converged state evaluates to a residual of 3.3×10^{-7} in this code’s assembler) and its converged fields to $\sim 10^{-10}$ relative L^2 . We document the governing equations and their nondimensionalization (including two deliberate idiosyncrasies of the reference run), the artificial viscosity and the smoothed density/pressure regularization that the continuation requires, the HDG formulation with static condensation, the damped-Newton globalization and the four-stage artificial-viscosity continuation, and the code layout, data formats, and verification gates.

Contents

1	Overview and scope	1
2	Physical problem	2
3	Governing equations	2
3.1	Conservative form	2
3.2	Sutherland viscosity	3
3.3	Artificial viscosity	3
3.4	First-order (mixed) form and the gradient sign convention	3
3.5	Density and pressure regularization (“floors”)	4
4	Nondimensionalization	5
4.1	The rounded deck constants (deliberately replicated)	5
5	Boundary conditions	5
6	HDG discretization	6
6.1	Mesh and geometry	6
6.2	Spaces and unknowns	6
6.3	Weak forms	6
6.4	Quadrature	7
6.5	Newton linearization and static condensation	7

7	Solution algorithm	8
7.1	Damped Newton (Exasim semantics, replicated exactly)	8
7.2	Linear solver (deliberate deviation)	8
7.3	Initialization modes	8
7.4	Artificial-viscosity continuation (M4)	8
7.5	Wall outputs	9
8	Code implementation	9
8.1	Layout	9
8.2	Selected implementation details	10
8.3	Verification gates	11
8.4	Running	11

1 Overview and scope

The target of this code is *replication*, not merely simulation: it reproduces a specific reference computation, the Exasim run archived at `Exasim-apps/runs/nsmach8-baseline`, with the same mesh (identical discrete geometry), the same physics model (transcribed term by term from the generated model code), the same HDG discretization (spaces, stabilization, quadrature), and the same nonlinear solver semantics (damped Newton with Exasim’s exact line-search and abort rules). The only deliberate deviation is the linear solver (a sparse direct factorization instead of Exasim’s preconditioned GMRES), which changes iteration counts but not the Newton fixed point.

Development was organized in four milestones. M1: physics/mesh/IO foundations verified against the Exasim data structures. M2: the HDG core verified on benign problems (manufactured solutions, freestream preservation, a Mach 4.5 sanity case). M3: the Mach 8 baseline replication from Exasim’s restart data. M4: a self-contained four-stage artificial-viscosity continuation from a damped-freestream initial condition, requiring no Exasim state data. All four milestone gates pass (§8.3).

2 Physical problem

The flow is two-dimensional, steady, laminar, compressible flow of a calorically perfect gas over the front half of a circular cylinder:

Mach number	$M_\infty = 8.03$
Reynolds number (based on cylinder radius)	$\text{Re} = 1.835 \times 10^5$
Prandtl number	$\text{Pr} = 0.71$
Specific-heat ratio	$\gamma = 1.4$
Freestream static temperature	$T_{\text{ref}} = 124.49 \text{ K}$
Wall temperature (isothermal)	$T_{\text{wall}} = 294.44 \text{ K}$
Sutherland constant	$T_s = 110.4 \text{ K}$

The domain is a half-annulus: the inner boundary is the cylinder surface (radius 1 after scaling), the outer boundary is a smooth arc standing off the bow shock, and the two straight segments on the $x = 0$ plane are supersonic outflow planes. A strong detached bow shock forms upstream of the cylinder (standoff ≈ 0.4 radii on the stagnation line); downstream of it the flow compresses to

$\rho \approx 5.6$ (inviscid Rankine–Hugoniot level), and in the cold-wall thermal boundary layer density rises further (the converged field reaches $\rho \approx 28$ at this Reynolds number). The steady solution is found directly by Newton’s method; there is no time stepping.

3 Governing equations

3.1 Conservative form

With conservative variables $\mathbf{u} = (\rho, \rho u, \rho v, \rho E)^T$, the steady compressible Navier–Stokes equations are

$$\nabla \cdot (F_{\text{inv}}(\mathbf{u}) - F_{\text{visc}}(\mathbf{u}, \nabla \mathbf{u}) - F_{\text{av}}(\nabla \mathbf{u})) = 0, \quad (1)$$

$$F_{\text{inv}} = \begin{pmatrix} \rho u & \rho v \\ \rho u^2 + p & \rho uv \\ \rho uv & \rho v^2 + p \\ \rho uH & \rho vH \end{pmatrix}, \quad F_{\text{visc}} = \begin{pmatrix} 0 & 0 \\ \tau_{xx} & \tau_{xy} \\ \tau_{xy} & \tau_{yy} \\ u\tau_{xx} + v\tau_{xy} + f_c \partial_x T & u\tau_{xy} + v\tau_{yy} + f_c \partial_y T \end{pmatrix}, \quad (2)$$

with $p = (\gamma - 1)(\rho E - \frac{1}{2}\rho(u^2 + v^2))$, total enthalpy $H = E + p/\rho$, and Stokes-hypothesis stresses

$$\tau_{xx} = \mu \frac{2}{3}(2u_x - v_y), \quad \tau_{xy} = \mu(u_y + v_x), \quad \tau_{yy} = \mu \frac{2}{3}(2v_y - u_x). \quad (3)$$

The “temperature” variable used throughout the model is the specific internal energy,

$$T = \frac{p}{(\gamma - 1)\rho} = e, \quad (4)$$

and the heat conduction coefficient is $f_c = \mu\gamma/\text{Pr}$, so that $f_c \nabla T = (\mu\gamma/\text{Pr})\nabla e$ equals the Fourier flux $(\mu c_p/\text{Pr})\nabla T_{\text{thermo}}$ in dimensional terms. This convention (temperature $\equiv$ internal energy) matters when comparing wall heat flux with other codes.

3.2 Sutherland viscosity

The dynamic viscosity follows Sutherland’s law, evaluated algebraically from the local state. Defining the nondimensional temperature ratio

$$T_r = \gamma M_\infty^2 \frac{p}{\rho} = \frac{T}{T_\infty}, \quad T_{\text{phys}} = T_{\text{ref}} T_r, \quad (5)$$

the code computes

$$\mu = \frac{1}{\text{Re}} T_r^{3/2} \frac{T_{\text{ref}} + T_s}{T_{\text{phys}} + T_s}, \quad T_s = 110.4 \text{ K}. \quad (6)$$

Note $T_r^{3/2}$: for $p < 0$ this is the square root of a negative number — the origin of the robustness problem addressed by the regularization of §3.5.

3.3 Artificial viscosity

Shock capturing uses a *frozen* Laplacian artificial viscosity: a fixed scalar field $a(\mathbf{x}) \geq 0$ adds isotropic diffusion to all four equations,

$$F_{\text{av}} = a(\mathbf{x}) \nabla \mathbf{u}, \quad (7)$$

i.e. the term $-\nabla \cdot (a \nabla \mathbf{u})$ on the left side of (1). The field a is *data*: it is never differentiated with respect to the state and never updated inside a Newton solve, so the exact Jacobian is preserved. Two sources are supported: the archived Exasim field (`vdg.bin`, which equals $0.025 \tanh(5(r-1))$ to 2.8×10^{-17}) and the analytic family

$$a(\mathbf{x}) = \lambda \tanh(c d(\mathbf{x})), \quad d(\mathbf{x}) = \sqrt{x^2 + y^2} - 1 \quad (8)$$

(d is the wall distance; exact for this geometry). The M4 continuation lowers (λ, c) through the schedule of §7.4.

3.4 First-order (mixed) form and the gradient sign convention

The HDG discretization rewrites (1) as a first-order system with an auxiliary gradient variable. Following Exasim’s convention *verbatim*,

$$\mathbf{q} = -\nabla \mathbf{u} \quad (\text{eight components in 2D}), \quad (9)$$

$$\mathbf{q} + \nabla \mathbf{u} = 0, \quad \nabla \cdot F(\mathbf{u}, \mathbf{q}) = 0. \quad (10)$$

All model formulas are then written in the “model-signed” gradient slots. For example, with $r_x \equiv -\partial_x \rho$ etc., the transcribed x -flux reads

$$f_x = \begin{pmatrix} \rho u + a r_x \\ \rho u^2 + p + \tau_{xx} + a (ru)_x \\ \rho uv + \tau_{xy} + a (rv)_x \\ \rho u H + u \tau_{xx} + v \tau_{xy} + f_c T_x + a (rE)_x \end{pmatrix}, \quad (11)$$

where $\tau_{xx}, \dots, T_x$ are built from the model-signed slots, so the *plus* signs in (11) are dissipative. Adopting this convention project-wide lets the generated Exasim model code (values and analytic Jacobians) be transcribed without any sign surgery; a dissipativity unit test (linearized entropy production ≤ 0) pins the convention.

3.5 Density and pressure regularization (“floors”)

The physics model exists in two variants, selected at run time (`physics.regularization` in the YAML deck):

Floor-free model (default). Equations (1)–(11) exactly as exported in the reference run’s `pdemodel.txt`. This is the model certified by the M3 parity gates.

Regularized model (floors). The model of the original `pdemodel_ns.m`, which clamps density and pressure through the smoothed hinge

$$\text{lmax}(x; \alpha) = x \left(\frac{\arctan(\alpha x)}{\pi} + \frac{1}{2} \right) - \frac{\arctan \alpha}{\pi} + \frac{1}{2}, \quad \alpha = 10^3, \quad (12)$$

which satisfies $\text{lmax}(x) \rightarrow x$ for $x \gg 1/\alpha$, $\text{lmax}(x) \rightarrow 0$ for $x \ll -1/\alpha$. Inside the flux only,

$$\rho \leftarrow \rho_{\min} + \text{lmax}(\rho - \rho_{\min}; \alpha), \quad \rho_{\min} = 10^{-2}, \quad (13)$$

$$p \leftarrow p_{\min} + \text{lmax}(p - p_{\min}; \alpha), \quad p_{\min} = 10^{-3}, \quad (14)$$

and the density and pressure gradient slots are scaled by the corresponding sensors (the derivative of the clamp),

$$d(x) = \frac{\arctan(\alpha\tilde{x})}{\pi} + \frac{\alpha\tilde{x}}{\pi(\alpha^2\tilde{x}^2 + 1)} + \frac{1}{2}, \quad \tilde{x} = (\text{clamped value}) - (\text{floor}), \quad (15)$$

applied as $(r_x, r_y) \leftarrow d_\rho(r_x, r_y)$ and $(p_x, p_y) \leftarrow d_p(p_x, p_y)$. The Sutherland law (6) is evaluated at the clamped (ρ, p) , so the flux is a smooth, globally defined function of the state — in particular finite at $p < 0$.

Two properties make this safe and necessary, respectively:

- **Inactive at converged states.** The clamp deviates from the identity by $O(1/(\alpha^3 x^2))$; at the converged states of this problem ($\min p \approx 0.0111 \gg p_{\min}$) the two models agree to $\sim 10^{-9}$ relative. All M3 parity gates are therefore blind to the difference, and both models share the same fixed points.
- **Load-bearing in the continuation.** Lowering the artificial viscosity from 0.06 to 0.04 sharpens the captured shock; the first Newton correction necessarily undershoots to $p \approx -0.009$ at the stagnation-line shock foot (both this code and Exasim’s own solver produce this trial state, agreeing to 0.02%). Without floors the trial residual is NaN (via $T_r^{3/2}$) and no line-search or pseudo-transient strategy can recover, because deciding to damp already requires evaluating the residual. With floors the same trial evaluates to a small finite residual, the step is accepted, and Newton restores positivity by itself within three iterations. The full forensic account is in `M4_FAILURE_REPORT.md`.

The regularized flux and its analytic Jacobian in `ns_physics.hpp` are mechanical transcriptions of symbolically generated code (`m4_diagnostic/my_model_floored.hpp`, produced by Exasim’s `text2code` from the floored model source); a unit test pins the transcription at exactly zero error over admissible *and* negative-pressure states.

4 Nondimensionalization

All quantities are scaled by freestream density ρ_∞ , freestream speed V_∞ , and the cylinder radius R :

$$\rho^* = \frac{\rho}{\rho_\infty}, \quad (u^*, v^*) = \frac{(u, v)}{V_\infty}, \quad p^* = \frac{p}{\rho_\infty V_\infty^2}, \quad E^* = \frac{E}{V_\infty^2}, \quad \mathbf{x}^* = \frac{\mathbf{x}}{R}, \quad \mu^* = \frac{\mu}{\rho_\infty V_\infty R}. \quad (16)$$

(The asterisks are dropped everywhere else in this document.) Consequently

$$\rho_\infty^* = 1, \quad (u, v)_\infty^* = (1, 0), \quad p_\infty = \frac{1}{\gamma M_\infty^2} = 0.011077\dots, \quad \mu_\infty^* = \frac{1}{\text{Re}}, \quad (17)$$

$$T_\infty = \frac{p_\infty}{(\gamma - 1)\rho_\infty} = \frac{1}{\gamma(\gamma - 1)M_\infty^2} = 0.0276937\dots, \quad (\rho E)_\infty = \frac{1}{2} + \frac{p_\infty}{\gamma - 1} = 0.5276937\dots \quad (18)$$

Physical temperature is recovered through $T_{\text{phys}} = T_{\text{ref}} T/T_\infty$ with $T_{\text{ref}} = 124.49$ K; the nondimensional isothermal-wall value is

$$T_{\text{isoW}} = T_\infty \frac{T_{\text{wall}}}{T_{\text{ref}}}. \quad (19)$$

4.1 The rounded deck constants (deliberately replicated)

The reference run’s parameter vector (in the code: `NSParams::mu[0..10]`) is, verbatim,

$$[\gamma, \text{Re}, \text{Pr}, M_\infty, \rho_\infty, (\rho u)_\infty, (\rho v)_\infty, (\rho E)_\infty, T_\infty, T_{\text{ref}}, T_{\text{wall}}] \\ = [1.4, 183500, 0.71, 8.03, 1, 1, 0, \mathbf{0.52769}, \mathbf{0.027694}, 124.49, 294.44].$$

The bold entries are *rounded literals*: the exact values are $0.5276937\dots$ and $0.0276937\dots$, so the deck is internally inconsistent at the 4×10^{-6} level. The model uses the two versions in different places, and this split must be replicated exactly (mixing them up costs 1.1×10^{-5} relative in the fields — enough to fail the parity gates):

- the **flux/Sutherland path** never reads the rounded T_∞ ; it forms $T_r = \gamma M_\infty^2 p / \rho$ algebraically, i.e. uses the full-precision $T_\infty = 1/(\gamma(\gamma - 1)M_\infty^2)$ derived from γ and M_∞ (`NSParams::TinfFlux()`);
- the **wall boundary condition** uses the rounded literal verbatim: $T_{\text{isoW}} = 0.027694 \times 294.44/124.49 = 0.06550101502128686$ (`NSParams::TisoW()`).

The freestream state used in Newton initializations is likewise the rounded $(1, 1, 0, 0.52769)$.

5 Boundary conditions

Boundary conditions are imposed *weakly through the trace unknown* $\hat{\mathbf{u}}$ (§6): on a boundary face the flux-continuity rows of the trace equation are replaced by residual rows $\oint_F \mu f_b \, ds = 0$, with f_b per boundary type ($\hat{u}_i$ denote trace components, u_i interior-state components):

Attribute	Boundary	Type	f_b
1 (21 faces)	cylinder $r = 1$	isothermal no-slip wall	$(u_0 - \hat{u}_0, -\hat{u}_1, -\hat{u}_2, \hat{u}_0 T_{\text{isoW}} - \hat{u}_3)$
2 (62 faces)	$x = 0$ plane	supersonic outflow	$\mathbf{u} - \hat{\mathbf{u}}$ (extrapolation)
3 (21 faces)	outer arc	supersonic freestream	$\mathbf{u}_\infty - \hat{\mathbf{u}}$ (Dirichlet on trace)

The wall rows extrapolate density, set zero velocity, and set trace internal energy to T_{isoW} (via $\hat{\rho E} = \hat{\rho} T_{\text{isoW}}$ at zero velocity). Two structural points, both verified against the generated Exasim code: the stabilization parameter τ does *not* appear in f_b ; and the *element* equation is unchanged on boundary faces (it keeps the interior numerical flux $\hat{f}$) — only the trace rows change. The energy wall row is bilinear in $\hat{\mathbf{u}}$, so its Jacobian carries both -1 and $+T_{\text{isoW}}$ entries.

6 HDG discretization

6.1 Mesh and geometry

The mesh is the reference run’s mesh: 651 curved quadrilaterals (31×21 , graded toward the wall), 704 vertices, 1,354 edges (104 on the boundary). Topology comes from `grid.bin`; the curved geometry is taken *exactly* from `xdg.bin`, the discrete Q_4 coordinate map on Exasim’s tensor Chebyshev–Gauss–Lobatto nodes (1D nodes $[0, (3 - \sqrt{5})/4, 1/2, (1 + \sqrt{5})/4, 1]$). Because a Q_4 polynomial re-expressed in a different nodal basis is exact, the MFEM $H^1 Q_4$ mesh nodes are set by evaluating Exasim’s basis at MFEM’s node locations; the resulting geometry reproduces `xdg.bin` to 1.1×10^{-15} . An analytic mesh generator (same 31×21 grading law: $\theta = 3\pi/2 - \pi\eta_2$, outer radius $4.7 + 1.7 \cos \theta$, radial stretching $\log \text{dec}(\eta_1, 5) = (1 - e^{-5\eta_1})/(1 - e^{-5})$) provides meshes for convergence studies and an independent M4 variant.

6.2 Spaces and unknowns

All spaces are *broken* (discontinuous) tensor-product polynomials of degree $p = 4$ on the reference square:

- $\mathbf{u}_h$: 4 components $\times$ 25 nodes per element ($651 \times 100 = 65,100$ unknowns);
- $\mathbf{q}_h$: 8 components $\times$ 25 nodes per element (never a solver unknown — eliminated exactly, see below);
- $\hat{\mathbf{u}}_h$: single-valued polynomial of degree 4 on every mesh edge (MFEM `DG_Interface_FECollection`), 4 components $\times$ 5 nodes $\times$ 1,354 faces = 27,080 unknowns — *the only globally coupled unknowns*. Boundary faces carry genuine trace unknowns (BCs are weak; nothing is eliminated strongly).

The trace space is stored with component-fastest ordering per face (20 contiguous scalar dofs per face), matching Exasim’s layout and giving clean 20×20 face blocks.

6.3 Weak forms

Let φ denote element test functions and μ face test functions. Per element K :

Gradient equation (linear, geometry-only). For each scalar component and direction d ,

$$\int_K \mathbf{q} \varphi = \int_K \mathbf{u} \partial_{x_d} \varphi - \oint_{\partial K} \hat{\mathbf{u}} n_d \varphi, \quad \implies \quad \mathbf{q}_d = C_d \mathbf{u} - E_d \hat{\mathbf{u}} \quad (20)$$

with precomputed 25×25 and 25×20 matrices $C_d = M^{-1} \int \partial_d \varphi \varphi$ and $E_d = M^{-1} \oint \varphi \varphi_{\text{face}} n_d$ (shared by all components; geometry-constant). $\mathbf{q}$ is eliminated through (20) from the start.

Element equation.

$$R_u = \int_K \nabla \varphi : F(\mathbf{u}, \mathbf{q}, a) \, d\mathbf{x} - \oint_{\partial K} \varphi \hat{f} \, ds = 0, \quad (21)$$

with the HDG numerical flux

$$\hat{f} = F(\hat{\mathbf{u}}, \mathbf{q}) \cdot \mathbf{n} + \tau (\mathbf{u} - \hat{\mathbf{u}}), \quad \tau = 1 \text{ (all equations, all faces)}. \quad (22)$$

The flux in (22) is the *full* (inviscid + viscous + AV) flux with the state slot replaced by the trace $\hat{\mathbf{u}}$ and the gradient slots kept from the element interior $\mathbf{q}$ — this specific evaluation point is load-bearing for both stability and Jacobian structure, and holds on boundary faces too.

Trace equation. On interior faces, flux continuity

$$\oint_F \mu (\hat{f}|_{K_1} + \hat{f}|_{K_2}) \, ds = 0 \quad (23)$$

(outward normals opposite); on boundary faces the rows are replaced by the weak boundary residuals of §5, mass-weighted with the same 5-point face rule. Because Exasim assembles the same weighted residuals, residual *norms* are comparable between the two codes, not just solutions.

6.4 Quadrature

Volume: 5×5 tensor Gauss–Legendre (degree 8, 25 points); faces: 5-point Gauss–Legendre. These match Exasim’s decoded rules exactly. Shape tables, geometry factors ($\det J$, $\text{adj } J$, face normals times surface Jacobians) and the frozen AV values at all volume/face quadrature points are tabulated once in the operator constructor (the mesh never moves).

6.5 Newton linearization and static condensation

Each Newton iteration assembles, per element, the exact-Jacobian blocks

$$A = \frac{\partial R_u}{\partial \mathbf{u}}, \quad B = \frac{\partial R_u}{\partial \mathbf{q}}, \quad F = \frac{\partial R_u}{\partial \hat{\mathbf{u}}}, \quad K = \frac{\partial R_h}{\partial \mathbf{u}}, \quad G = \frac{\partial R_h}{\partial \mathbf{q}}, \quad H = \frac{\partial R_h}{\partial \hat{\mathbf{u}}}, \quad (24)$$

folds the gradient dependence through (20) ($A += \sum_d B_d C_d$, $F -= \sum_d B_d E_d$, and likewise for K, G, H), and condenses:

$$H_c = H - K A^{-1} F \quad (80 \times 80), \quad b_c = K A^{-1} R_u - R_h. \quad (25)$$

Eliminating $\mathbf{q}$ first reduces the per-element factorization from 300×300 to 100×100 (a $27\times$ flop reduction); peak solver memory is ≈ 106 MB. The condensed system $H_c^{\text{glob}} \Delta \hat{\mathbf{u}} = b_c^{\text{glob}}$ is a $27,080 \times 27,080$ sparse matrix with ≤ 140 nonzeros per row (each face couples to itself and at most six sibling faces). After the trace solve, the local recovery is

$$\Delta \mathbf{u} = -A^{-1}(R_u + F \Delta \hat{\mathbf{u}}), \quad \text{then} \quad \mathbf{q} := C \mathbf{u} - E \hat{\mathbf{u}} \quad (\text{full recompute}). \quad (26)$$

7 Solution algorithm

7.1 Damped Newton (Exasim semantics, replicated exactly)

Per iteration: full reassembly at the current $(\mathbf{u}, \mathbf{q}, \hat{\mathbf{u}})$; direct solve of the condensed system; update $\hat{\mathbf{u}} += \alpha \Delta \hat{\mathbf{u}}$ with local recovery of $\mathbf{u}$ and recomputation of $\mathbf{q}$; then the line search. The line-search and abort rules are transcribed from Exasim’s `NewtonSolver`:

- trial $\alpha = 1$; **while** the residual increased *and* $\alpha > 0.1$: halve α and retry. Because the guard tests α *before* halving, the admissible steps are $\alpha \in \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$ and the step $\alpha = 0.0625$ is *accepted even if the residual still increased* — a quirk that is load-bearing at Mach 8;
- hard abort only if the trial residual is NaN, or increased *and* exceeds 10^6 ;
- convergence when $\|R_u\| + \|R_h\| < 10^{-6}$ (stacked-norm definition, matching Exasim); at most 30 iterations.

7.2 Linear solver (deliberate deviation)

The condensed systems are solved by a sparse direct factorization (MUMPS via PETSc, `KSPPREONLY + PCU`). Exasim uses restarted GMRES with a degree-20 polynomial preconditioner; that choice affects inner iteration counts only, not the Newton fixed point, and is deliberately not replicated. An optional GMRES + face-block-Jacobi mode exists for solver studies. A pseudo-transient continuation hook (backward-Euler term $(1/\Delta\tau)$ added to A , SER ramp) is wired but *off* in all passing configurations — the reference schedule needs no PTC.

7.3 Initialization modes

- **udg_file** (M3): load $\mathbf{u}$ from Exasim’s `udg.bin` (components 0–3 only), set $\hat{\mathbf{u}}$ to the one-sided trace of $\mathbf{u}$ (from the lower-index element on each face, matching Exasim’s restart), then recompute $\mathbf{q} = C\mathbf{u} - E\hat{\mathbf{u}}$. The stored gradient slots are deliberately discarded, exactly as Exasim’s solver does on restart.
- **damped_freestream** (M4): the `pdeapp_ns.m` initial condition

$$\rho = 1, \quad \rho u = \tanh(10d), \quad \rho v = 0, \quad \rho E = T_\infty \left[\left(\frac{T_{\text{wall}}}{T_{\text{ref}}} - 1 \right) e^{-10d} + 1 \right] + \frac{1}{2} \tanh^2(10d), \quad (27)$$

with $d = r - 1$ the wall distance: freestream flow smoothly brought to rest at the wall with a near-wall thermal layer consistent with the hot wall.

7.4 Artificial-viscosity continuation (M4)

The self-contained solve replicates the reference script’s four steady solves with frozen AV fields, each warm-starting from the previous:

$$a_k(\mathbf{x}) = \lambda_k \tanh(c_k d), \quad (\lambda_k, c_k) \in \{(0.06, 30), (0.04, 30), (0.025, 30), (0.025, 5)\}. \quad (28)$$

At each stage start the trace is re-initialized to the one-sided interior trace and $\mathbf{q}$ recomputed (Exasim’s per-stage restart semantics; this makes the per-stage residual histories directly comparable between the codes). The regularized model of §3.5 is required from stage 2 on; with it, the run needs no PTC, no extra stages, and no damping after stage 1:

Stage	AV field	Newton updates	Damped steps	Final residual
1	$0.06 \tanh(30d)$	9	5 (min $\alpha = 0.125$)	2.52×10^{-9}
2	$0.04 \tanh(30d)$	4	0	2.30×10^{-8}
3	$0.025 \tanh(30d)$	4	0	2.49×10^{-7}
4	$0.025 \tanh(5d)$	4	0	3.30×10^{-7}

The 9/4/4/4 signature matches Exasim’s own (floored) runs of the same schedule; the stage-2 histories track step for step, both accepting a first trial whose minimum pressure is ≈ -0.0090 (absorbed by the floors). The end state matches the M3 replication to 3.4×10^{-6} in wall heat flux — far inside the 1% acceptance gate.

7.5 Wall outputs

Wall postprocessing evaluates, at the 5 Gauss points of each wall face and with the trace state in the state slot and interior $\mathbf{q}$ in the gradient slots (exactly the reference `Fint` functional),

$$q_w = f_c (T_x n_x + T_y n_y) + \tau (\rho E - \widehat{\rho E}), \quad C_p = 2 (p(\hat{\mathbf{u}}) - p_\infty), \quad (29)$$

the latter because the dynamic pressure is $\frac{1}{2} \rho_\infty V_\infty^2 = \frac{1}{2}$ in these units. The shock standoff is located as the maximum of $|\partial_x \rho|$ along the stagnation line from 1,000 samples.

8 Code implementation

8.1 Layout

The app is a flat directory of small translation units over MFEM 4.8.1-dev (MPI + METIS + PETSc/MUMPS build; serial execution — 651 elements is decisively serial territory):

File	Contents
<code>ns_physics.hpp</code>	Header-only physics on raw <code>double</code> arrays (no MFEM types; shared with the planned DG driver): fluxes, Sutherland, boundary residuals, analytic Jacobians transcribed from the generated Exasim model; the regularized (floored) flux/Jacobian pair behind <code>NSParams::regularized</code> ; sign-flip adapters for physical-gradient callers.
<code>exasim_io</code>	Exasim binary readers (3-double header + column-major payload), Chebyshev–Gauss–Lobatto basis evaluation, exact 25×25 change-of-basis, per-rank <code>elem_part</code> parsing for the partitioned reference outputs.
<code>exasim_mesh</code>	Mesh converter (<code>grid.bin</code> topology + <code>xdg.bin</code> exact geometry + boundary attribution by the run’s own predicates) and the analytic mesh generator.
<code>hdg_ns_operator</code>	The HDG core: precomputed tables, C_d/E_d , residual and Jacobian assembly, boundary-row substitution, static condensation into an MFEM <code>SparseMatrix</code> , local recovery, residual-only path, diagnostics (min ρ/p , symmetry error).
<code>hdg_newton.hpp</code>	Damped Newton of §7.1 with the PETSc direct solve and the optional PTC hook; failures are reported, not thrown.
<code>wall_post</code>	C_p , <code>Fint</code> heat flux, shock standoff, and the field/wall comparison machinery used by the M3/M4 gates.
<code>hdg_ns_driver.cpp</code>	YAML-configured driver (mesh source, physics, AV mode, init mode, Newton/PTC, continuation schedule, outputs); ParaView output of conservative/primitive/AV fields.
<code>test.*</code>	The verification suite (§8.3).
<code>m4_diagnostic/</code>	Forensic artifacts of the M4 investigation: the floored model source and its generated code (the transcription reference), stage-driving script, logs, and the first-failing-step dumps.

8.2 Selected implementation details

- **Exact geometry and state transfer.** Both the Q_4 geometry and all state I/O go through exact polynomial change-of-basis between Exasim’s Chebyshev–Lobatto nodes and MFEM’s node sets (round-trip error $\leq 7 \times 10^{-14}$); naive nodal copying would be $O(10^{-3})$ wrong at $p = 4$.
- **Jacobians are transcribed, not derived.** Both model variants’ analytic Jacobians come from Exasim’s symbolic code generator; unit tests compare the transcriptions against lightly adapted copies of the generated headers at 10^{-13} (the floored pair matches *exactly*), plus

central finite-difference checks. Hand-differentiation of the Sutherland/ sensor chain rules was deliberately avoided.

- **Sparsity with explicit zeros.** The condensed matrix graph is built once with `skip_zeros=0`; exact zeros are guaranteed in boundary blocks and dropping them would corrupt later re-assemblies.
- **Frozen-AV tabulation.** $a(\mathbf{x})$ is tabulated once at all volume and face quadrature points; it enters the Jacobian only as $\partial f_{c,d}/\partial q_{c,d} += a$ (identity blocks scaled by data).
- **Face transformations.** Face physical coordinates are always evaluated through the face transformation objects; MFEM reuses mutable transformation storage, and retaining stale element pointers across face queries corrupts face data (found the hard way in M2).
- **Build hygiene.** Every translation unit that includes `ns_physics.hpp` lists it as a makefile dependency; `NSParams` is passed across TU boundaries, and a stale-object ABI mismatch after the struct grew produced a heap abort before the dependencies were fixed.
- **Timings** (single process, development host): a full M3 Newton solve (4 iterations) spends ≈ 5.1 s in assembly and ≈ 1.1 s in the direct trace solves; each M4 stage runs in ≈ 7 –14 s.

8.3 Verification gates

`make test` runs the milestone gates in sequence:

Gate	Statement	Result
M1 physics	flux/BC values and all Jacobians vs the generated model at 1000 random states; FD consistency; dissipativity sign	$\leq 10^{-13}$; 4×10^{-9}
M1 regularized	floored flux/Jacobian vs generated floored model over admissible + floor-straddling (incl. $p < 0$) states; FD; floored/unfloored agreement at admissible states; M4 failure state finite (floored) and NaN (unfloored)	exactly 0; 4.9×10^{-7} ; 4.2×10^{-7} ; pass
M1 mesh/IO	topology/attributes; exact <code>xdg</code> geometry; change-of-basis round trips; quadrature identity	1.1×10^{-15}
M2 core	trace-orientation consistency; global condensed Jacobian vs directional FD; MMS convergence order ≥ 4.8 ; freestream preservation; $M=4.5$ sanity case	pass (9.7×10^{-10} FD)
M3 replication	residual of Exasim's converged state in this assembler; Newton from <code>udg.bin</code> ; field L^2 vs Exasim; wall C_p /heat flux/standoff	3.3×10^{-7} ; $\sim 10^{-10}$; $\sim 10^{-9}$
M4 continuation	four-stage schedule (no PTC) on converted and analytic meshes; final wall heat flux and standoff vs M3 within 1%	9/4/4/4; 3.4×10^{-6}

The strongest single test is the M3 residual cross-check: Exasim's converged $(\mathbf{u}, \mathbf{q}, \hat{\mathbf{u}})$, loaded into this code's assembler, must evaluate to a residual at Exasim's own convergence tolerance. It certifies discretization parity independent of any solver behavior, and it gates everything downstream.

8.4 Running

```
make                # builds driver + all test binaries
make test          # M1-M4 verification gates
./hdg_ns_driver -i Input/input_mach8_baseline.yaml      # M3 replication
./hdg_ns_driver -i Input/input_mach8_continuation.yaml  # M4 continuation
```

A deck selects the mesh source (`exasim` converter or `analytic`), physics constants and regularization, AV mode (`file` or `tanh`), initialization, Newton/PTC settings, the continuation schedule, and output paths. Outputs are high-order ParaView collections (conservative/primitive/AV fields), wall CSVs (θ, x, y, C_p, q_w) at wall Gauss points, and per-stage continuation history/summary CSVs.

Further reading in the repository

- `PLAN_HDG.md` — the full implementation plan, with file/line references into the Exasim sources for every discretization decision.
- `M4_FAILURE_REPORT.md` — the continuation failure forensics: the negative-pressure mechanism, the cross-code first-failing-step comparison, and the floored-model reconstruction that resolved it.
- `NOTES.md` — milestone-to-milestone findings (data-format gotchas, orderings, measured parity numbers).
- `PLAN_DG.md` — the planned sibling nodal-DG driver sharing `ns.physics/exasim.io/exasim.mesh`.